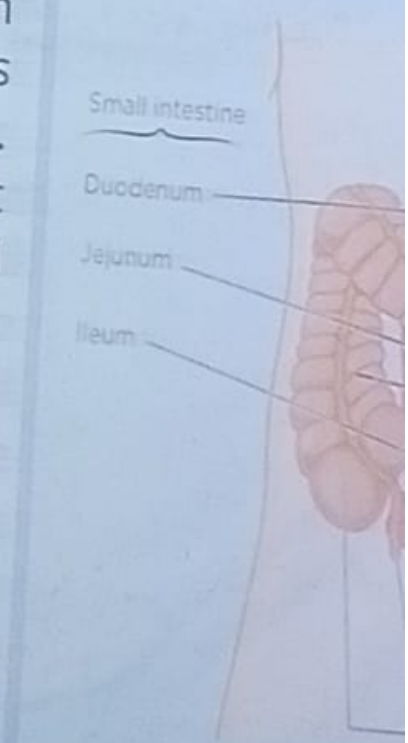
Duodenum, jejunum and ileum constitute the small intestine. Small intestine commences from the distal part of stomach and ends at the ileocecal junction. Ironically, it is about five times longer than the large intestine. But because it has a smaller diameter than the large intestine (about 2.54cm versus 7.62cm) it is called 'small'. It comprises the duodenum (25cm), jejunum (around 2.5m) and ileum (around 3.5m).

The jejunum and the ileum are attached to the posterior wall of the abdomen by the mesentery making their entire convolution freely mobile in the abdominal cavity.

Jejunum is the middle section of the small intestine, while the ileum is the final section. They are responsible for absorbing nutrients from digested food, with the jejunum primarily absorbing most nutrients and the ileum focusing on vitamin B12 and bile acids.

Jejunum: Jejunum is the middle part of the small intestine situating between the duodenum and the ileum. The jejunum is approximately 2.5 meters long. The jejunum begins at the **duodenojejunal flexure.** There is however, no clear external demarcation between the jejunum and ileum – although the two parts are macroscopically different. The ileum ends at the **ileocecal junction.**

Fig 1 Anatomy of the sm



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Fig. Anatomy of the small and large intestines



Distinguishing features of jejunum from ileum:

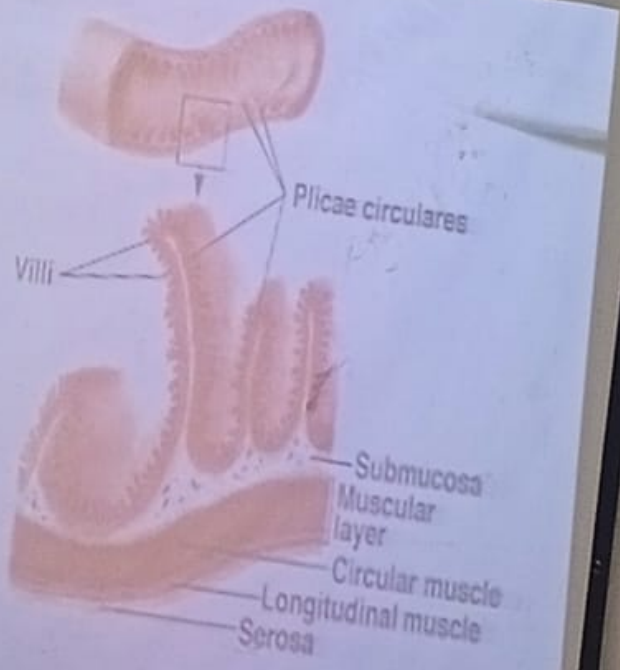
Wall thickness: The jejunum has a thicker wall compared to the ileum.

- **Vascularity:** The jejunum is more vascularized, appearing a deeper red colour compared to the ileum.
- **Mesenteric fat:** The ileum has more mesenteric fat surrounding it compared to the jejunum.
- **Peyer's patches:** Peyer's patches, which are lymphoid tissue aggregates, are primarily found in the ileum.

Microscopic features:

Plicae circulares (valves of Kerckring):

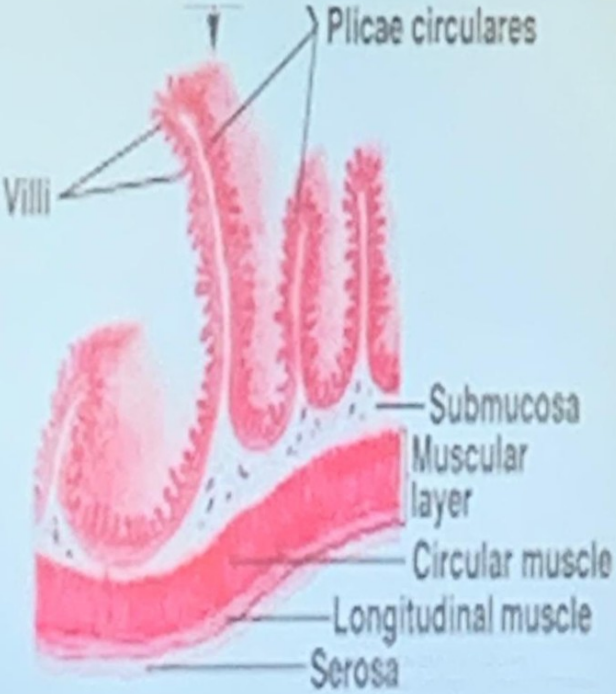
They are permanent, circular folds of mucous membrane found in the small intestine, primarily in the jejunum. The circular folds in the intestinal lining increases the surface area for absorption. They do not disappear even when the intestine distends.



Villi: Tiny finger-like projections on the mucosal surface that further enhance nutrient absorption.

It tends to be flatter in duodenum but longer and more finger-like in jejunum.

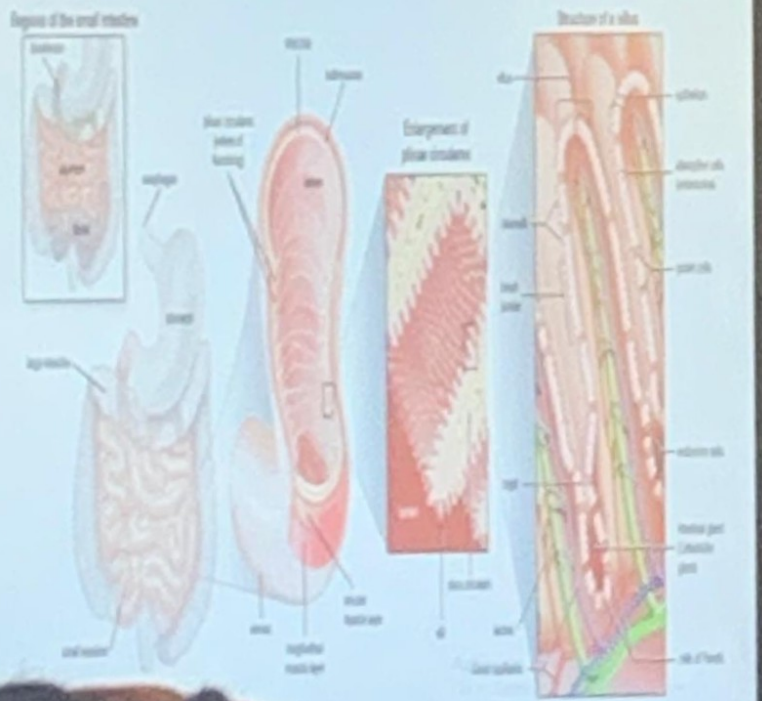
Primarily absorbs most of the nutrients from digested food.



CLEVERTOUCH

The ileum: It is the longest part of the small intestine, over 3 meters in length making up about three-fifths of its total length. It is thinner and less vascular than the jejunum, and the circular folds are less dense and more separate. At the distal end, the ileum is separated from the large intestine by the ileo-caecal valve, a sphincter formed by the circular muscle layers of the ileum and caecum, and controlled by nerves and hormones. The ileo-caecal valve prevents reflux of the bacteria-rich content from the large intestine into the small intestine.

Ileum is rich in immune tissue (lymphoid follicles). A characteristic feature is Peyer's patches, found lying in its mucosa, which is an important feature of that part of small intestine. One Peyer's patch is around 2-5cm long and consists of around 300 aggregated lymphoid follicles and is concentrated in the distal ileum and serve to keep bacteria from entering the blood stream.

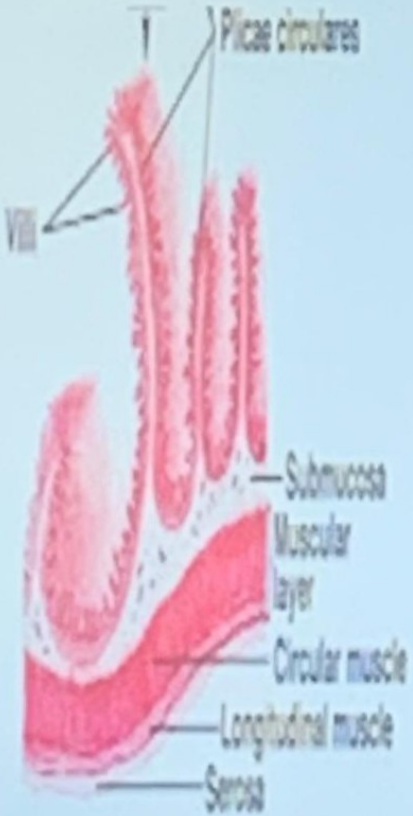


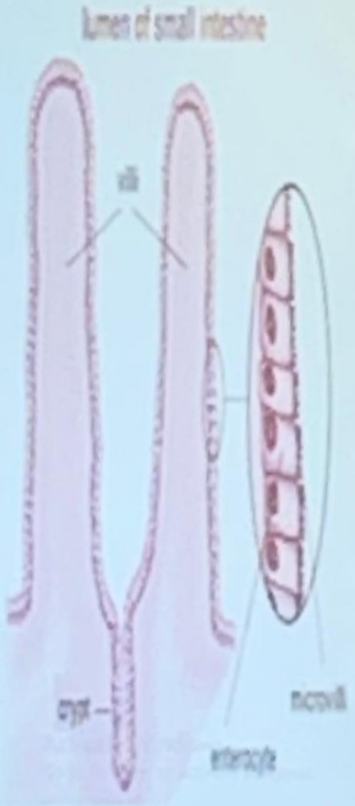
Microscopic features of ileum:

Plicae circulares (valves of Kerckring):

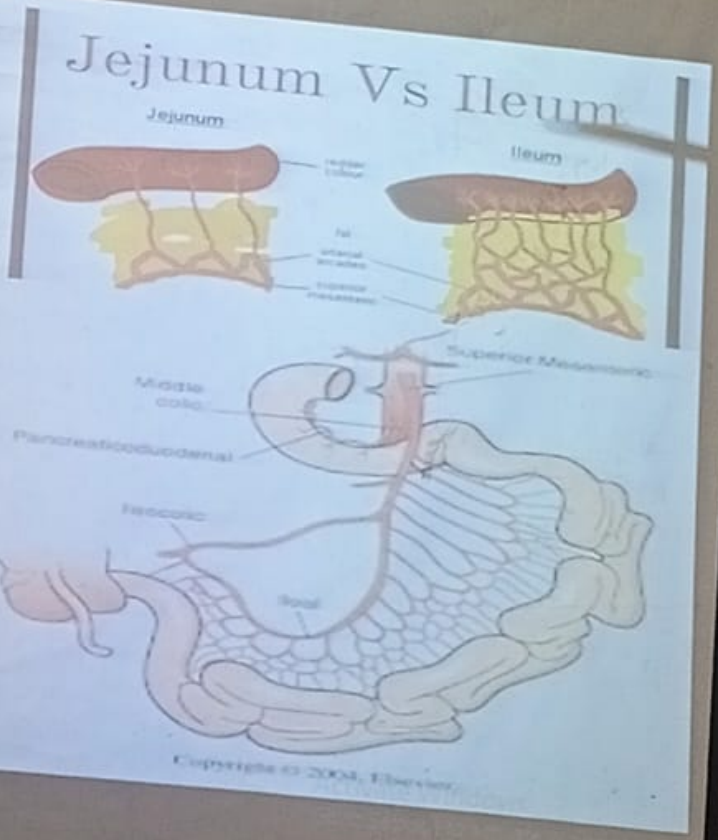
The circular folds are less dense and more separate.

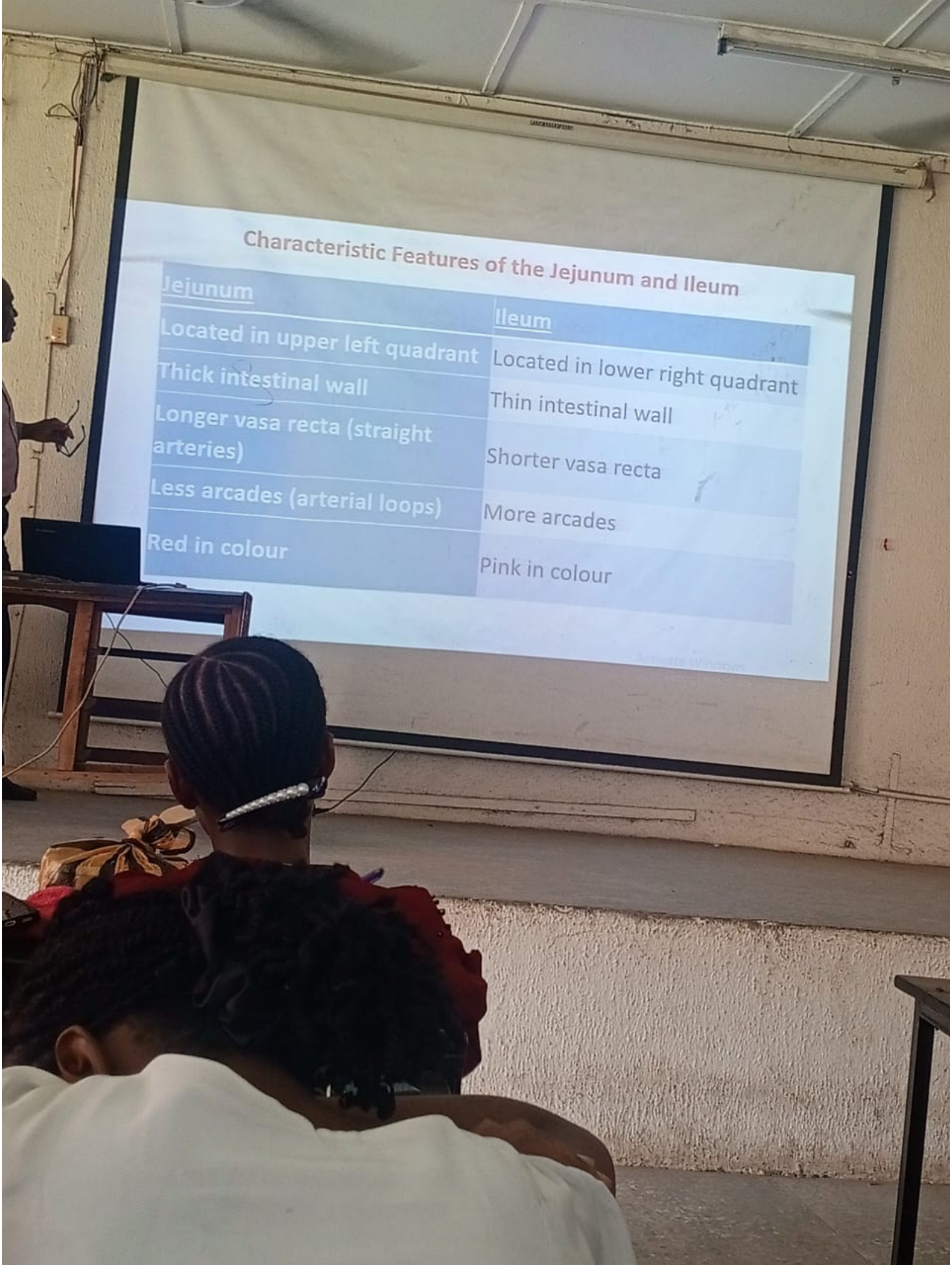
Villi: It is less prominent. However, they are covered with smaller hair-like structures called microvilli. Its characteristic feature is Peyer's patches, found lying in its mucosa.





Blood supply. The superior mesenteric artery arises from the aorta at the level of the L1. They pass between the two leaves of mesentery. These branches anastomose to form loops, called **arcades**. From the arcades, long and straight arteries arise, called **vasa recta**.




Characteristic Features of the Jejunum and Ileum
Jejunum
Ileum
Located in upper left quadrant
Located in lower right quadrant
Thick intestinal wall
Thin intestinal wall
Longer vasa recta (straight arteries)
Shorter vasa recta
Less arcades (arterial loops)
More arcades
Red in colour
Pink in colour

1. **Genetic Factors**
   - Family History
   - Inherited Syndromes
   - Gene Mutations.
2. **Chronic Inflammatory Conditions**
   - Ulcerative Colitis & Crohn's Disease
3. **Dietary & Lifestyle Factors**
   - High Consumption of Red & Processed Meat
   - Low Fiber Diet
   - Obesity & Sedentary Lifestyle
   - Smoking & Heavy Alcohol Consumption
4. **Metabolic & Microbiome Factors**
   - Type 2 Diabetes
   - Gut Microbiome Dysbiosis
5. **Environmental & Medical Factors**
   - Carcinogen Exposure
   - Previous Radiation Therapy

